Undertow - Step 4: the bot, Protocol view and watchlist

Built and tested on 10 October 2026. The bot runs in paper mode only; nothing here has traded real money.

What was added

Piece	Status
Market-making bot (`undertow/bot.py`)	Built. **Paper mode** works and ran on live data. The live executor is written but **locked and never run against the chain**.
Bot tab on the dashboard	Built. Shows mode, equity, position, quotes and recent events.
Protocol tab	Built, partly. Open interest, accounts, balances, per-market positions and collateral. No volume, fees, deposits or liquidation history (needs event history).
Watchlist and compare (Wallets tab)	Built. Save wallets, see them side by side. Saved in your browser only.

How to run

```
python -m undertow.server --refresh
python -m undertow.bot --paper --cycles 10 --interval 45
```

Keep the server running: the bot reads the data it keeps fresh, and pulls its quotes if the data gets older than two minutes. Open the **Bot** tab at http://localhost:8000 to watch it.

What the bot does

Every cycle it keeps one buy order slightly below Perpl's mark price and one sell order slightly above, and earns the gap when both fill. It only places **post-only** orders, so it always pays the low maker fee (0.0045%) and never the high taker fee.

Rule	Setting	What happens
Gap	8 bps each side	Quotes sit this far from the mark price. The round-trip fee is under 1 bp, so the gap is the profit.
Order size	$20 per side	Small on purpose, for a $50 to $100 account.
Inventory lean	up to 6 bps	If it holds a long, both quotes shift down so it is more likely to sell (and the reverse).
Position limit	$60	It stops quoting the side that would grow the position beyond this.
Zone near	zone of $25,000+ within 1.5%	Gap widens x2.5.
Zone near and book thin	zone value over 2.0x resting orders	**Pulls all quotes.** Uses the Radar's zone data.
Stale data	older than 120s	Pulls quotes.
Daily loss	$5	**Halts for good** and sends an alert.
Leverage	above 2x	**Halts.**
Own liquidation	closer than 25%	**Halts.**
Kill switch	a file named KILL in data/bot	**Halts** on the next cycle.

Built-in safety

- **Every order expires by itself** about 60 seconds after it is placed (150 blocks). If the program crashes or your laptop sleeps, the orders vanish without any cancel command.
- **Every order has an increasing request number**, which the exchange treats as a duplicate guard, so the same order is never applied twice.
- **The private key is never put on a command line.** The tool reads it from an environment variable, and a test checks that no command contains it.

- **Live mode is locked.** It refuses to start unless a key is set in the environment and a second switch (UNDERTOW_ALLOW_LIVE=yes) is on. Neither is set. Two tests check the refusal.
- **Alerts** print to the screen and go to Telegram if you set a bot token and chat number. Telegram sending was not tested (no token).

What the paper run actually showed

A real run on live Perpl data: 6 cycles, 45 seconds apart, BTC. Quotes are computed from the real mark price.

Cycle	Mark	Bid	Ask	Bid to ask (bps)	Action
1	82,456.2	82,390.2	82,522.2	16.0	quote
2	82,456.5	82,390.5	82,522.5	16.0	quote
3	82,455.5	82,389.5	82,521.5	16.0	quote
4	82,455.5	82,389.5	82,521.5	16.0	quote
5	82,470.6	82,404.6	82,536.6	16.0	quote
6	82,499.9	82,433.9	82,565.9	16.0	quote

Result: 0 fills, profit and loss $0.00. The BTC mark moved from 82,455.5 to 82,499.9 (about 5.4 bps) during the run, less than the 8 bps each-side distance from the mark to each quote, so no simulated order was reached. This shows that the quoting works on real data. It is **not** evidence that the bot makes money.

How well is it tested?

- **48 automated tests pass** (20 for the bot). They cover: quotes straddle the mark and respect price and size steps; the gap beats fees; inventory leaning; position limit; stale data; kill switch; daily loss, leverage and liquidation-distance halts; a halt staying halted; big zone with a thin book pulls; big zone with a deep book only widens; small or distant zones ignored; simulated fills and a full round trip with fees; the live lock; and the shape of the order command.
- The fill model is crude: an order counts as filled when the mark price crosses it, checked once per data refresh (about every 45 seconds).
- **Not tested at all:** placing a real order on any network, Telegram sending, and cancelling by order number (orders rely on their expiry instead).

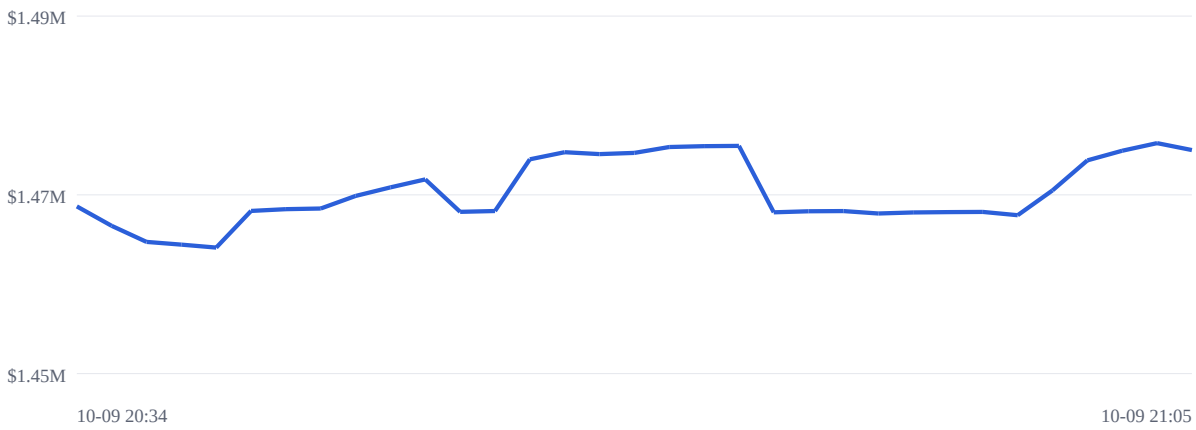
Settings are guesses

The gap, order size, limits and zone thresholds above are my starting values, not tuned on data. The bot also only knows BTC's price step (0.1) and size step (0.00001), which I read from the live order book. Other markets need their own values.

Protocol view

Exchange-wide numbers from the live data. Open interest now: **$1.47M** across 433 accounts and 652 positions. The accounts holding positions have $1.82M in balances.

Open interest since collection began (33 snapshots from 2026-10-09 20:34:22 UTC):



Market	Open interest	Share	Longs	Shorts	Long collateral	Short collateral
BTC	$677k	45.9%	141	90	$120k	$114k
ETH	$436k	29.6%	40	40	$121k	$80k
SOL_v2	$121k	8.2%	28	25	$21k	$39k
MON	$65k	4.4%	73	17	$25k	$37k
ZEC	$57k	3.9%	25	11	$14k	$14k
PUMP	$39k	2.6%	14	10	$7,468	$16k
HYPE	$38k	2.6%	37	19	$8,248	$6,813
LIT	$19k	1.3%	21	5	$8,934	$6,434

What this view is missing

- **Trading volume, fees, deposits and withdrawals, liquidation history.** These need every past event, which needs the Envio indexer. The Perpl Risk Tool bounty lists several of these, so this view does **not yet meet that bounty**.
- **Open-interest history only goes back to when the refresher first ran** (a few hours of snapshots), not to the exchange's launch.
- Funding rate is shown as the contract reports it; I have not verified its time unit.

Watchlist and compare

On the Wallets tab, press Watch on any wallet. Saved wallets appear side by side with balance, position value, profit or loss, highest leverage, nearest liquidation and how many positions have other traders' zones in their path. It is stored in your browser only, so it is not shared or synced.

What is left, and what needs you

Item	What it is	Status
Envio indexer	Needed for volume, fees, liquidation history, backtesting the cascade model, and the $1,000 Envio prize	Needs your choice: Docker + Envio, or Envio hosted
Bot on Perpl's test network, then mainnet	Needs a separate wallet with funds and its key in an environment variable on your machine	Needs you. I will not handle keys or funds
Bot tuning	Settings are untuned guesses; needs real fills or an event replay	After testnet
Wallet statistics	Win rate, profit factor, biggest fall, streaks, holding time	Needs trade history (Envio)
Order form with risk preview	Shows your liquidation, zones in your path, lower-leverage option, funding cost per day	Not started
Funding cost tracker	Cost growing block by block	Not started; needs funding unit confirmed
Liquidation-aware orders	Cancel or shrink an order when a zone gets close	Not started; needs a faster feed than 45 s
Remaining correctness checks	24h volume, own test account, order book vs Perpl's screen	Volume needs Envio
Public link, logo, GitHub, four videos, README	Track 1 deliverables	Not started

Schedule. The deadline is 13 October. The two items that decide whether the bot counts as 'real on-chain activity' (a funded wallet and days of live trading) both need you and time that cannot be compressed. If the wallet is funded today, the bot gets about three days of activity at best.